



Free MCAT Practice Set

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Answer key and worked explanations start on the next page. Independent study material, not affiliated with the AAMC.

Q1 CHEM/PHYS

An ambulance siren emits sound at a constant, fixed frequency as it drives toward a stationary pedestrian, then passes the pedestrian and drives away. Which statement best describes what the pedestrian actually hears?

- A A frequency lower than the emitted frequency while approaching, and higher than emitted while receding
- B A constant frequency the entire time, with only the loudness changing as the ambulance passes
- C A frequency higher than the emitted frequency while approaching, dropping to lower than emitted once the ambulance is moving away
- D The siren's true emitted frequency itself increases as the ambulance approaches and decreases as it recedes

Q2 CHEM/PHYS

A chain contains three stereocenters, and no part of the molecule is symmetric enough to create a meso form. How many total stereoisomers exist, and how many enantiomeric pairs does that represent?

- A 8 stereoisomers, 4 enantiomeric pairs
- B 6 stereoisomers, 3 enantiomeric pairs
- C 4 stereoisomers, 2 enantiomeric pairs
- D 8 stereoisomers, 2 enantiomeric pairs

Q3 CHEM/PHYS

Two secondary alkyl halides, one with iodine and one with fluorine as the leaving group, are each dissolved separately in a weakly nucleophilic, polar protic solvent with no added base. Which reacts faster by the S_N1 pathway, and why?

- A The fluoride, because fluorine's small size lets the nucleophile approach more easily
- B They react at the same rate, because S_N1 rate depends only on carbocation stability, not on the leaving group
- C The iodide, because the C–I bond is weaker and iodide is a much better leaving group, more easily accommodating negative charge as it departs
- D The fluoride, because C–F bond polarity makes the carbon more electrophilic

Q4 BIO/BIOCHEM

Based on side-chain chemistry alone, which of the following amino acids is most likely to be found buried in the hydrophobic core of a folded globular protein?

- A Serine
- B Aspartate
- C Leucine
- D Lysine

Q5 BIO/BIOCHEM

An alpha helix within a single polypeptide chain is a classic example of secondary structure. Which type of interaction is primarily responsible for stabilizing this level of protein structure?

- A Hydrogen bonds between backbone carbonyl and amide groups
- B Peptide bonds between adjacent amino acid side chains
- C Disulfide bridges between cysteine residues
- D Ionic attractions between oppositely charged side chains

Q6 PSYCH/SOC

Every spring, residents of a small town gather for a week-long festival featuring shared meals, parades, and ceremonies honoring the town's history. A sociologist argues that the festival endures because it reaffirms shared values, strengthens social bonds, and helps the community remain stable and cohesive. This analysis best reflects which theoretical perspective?

- A Conflict theory
- B Symbolic interactionism
- C Functionalism
- D Rational choice theory

Q7 PSYCH/SOC

Decades ago, a city's wealthiest property owners wrote zoning rules that steered factories and waste facilities into the neighborhoods where the poorest residents lived, and those rules still determine where pollution is concentrated today. A sociologist argues that this arrangement persists because it protects the interests of powerful groups at the expense of groups with fewer resources. This analysis best reflects which theoretical perspective?

- A** Conflict theory
- B** Functionalism
- C** Symbolic interactionism
- D** Social exchange theory

PASSAGE

A London theatergoer of 1790 inhabited a house that never went dark. Patrons ate, drank, and conversed through the performance; they demanded that favorite arias be repeated on the spot and hissed actors who displeased them; the pit talked back to the stage as a matter of right. A century later the same institutions had been transformed. Auditoriums were darkened, latecomers were locked out until intermission, and applause was confined to sanctioned pauses. The audience had learned, above all, to be silent. The conventional explanation treats this change as refinement: audiences grew more educated, appreciation deepened, and reverent silence followed as naturally as quiet follows concentration. Revisionist historians tell a less flattering story. Silence, they argue, was manufactured. Theater managers and critics campaigned against vulgar participation; etiquette manuals instructed the aspiring middle class in the new discipline; rising ticket prices filtered the house. Wagner, who insisted that the auditorium at his festival theater in Bayreuth be darkened so that nothing would compete with the stage, was not responding to a refined public; he was engineering one. On this account, silence did not express a deepened love of art. It redefined art as something to be revered rather than used, and it converted the concert hall into a place where a class could display its self-command. The revisionist case is largely persuasive, but it can be pressed too far. The repertory itself was changing: a symphony that unfolds over forty minutes rewards sustained attention in a way that a string of familiar airs does not, and some of the new discipline surely tracked the new demands of the music. Nor was silence purely imposed from above; workingmen's concert societies adopted the same codes without any manager compelling them. What the period's history actually shows is neither progress nor conspiracy but something more interesting: that how to attend is itself an artifact with a history. The silent listener hears more of the score and less of her neighbors, and whether that exchange counts as gain or loss is a question the nineteenth century answered for us before we could ask it.

Which of the following best expresses the central thesis of the passage?

- A Nineteenth-century audiences came to appreciate art more deeply than their predecessors, and reverent silence was the natural result of that deepened appreciation.
- B Audience silence was imposed entirely by managers and critics as an instrument of class distinction, with no contribution from changes in the music itself.
- C Conventions of audience behavior are historical products, and the rise of silent listening reflected both deliberate discipline and a changing repertory rather than simple refinement.
- D The transformation of theater audiences demonstrates that changes in artistic taste are ultimately driven by ticket prices and other economic barriers to attendance.

Answer Key & Explanations

Don't just check right or wrong. Read why the trap answers are tempting, that's where the points hide.

Q1 C CHEM/PHYS

The Doppler effect describes how a moving source changes the frequency perceived by a stationary observer, even though the source's true emitted frequency never changes. As the ambulance approaches, it is chasing its own emitted wavefronts, compressing them ahead of it, so the pedestrian intercepts more wave cycles per second than the siren actually emits, perceiving a higher pitch. Once the ambulance passes and moves away, it is now outrunning wavefronts headed back toward the pedestrian, stretching them out, so the pedestrian perceives a lower pitch than the true emitted frequency. This produces the familiar downward 'pitch drop' as an ambulance passes.

Q2 A CHEM/PHYS

With three stereocenters and no internal symmetry to create a meso relationship, the 2^n rule applies without correction: $2^3 = 8$ distinct stereoisomers exist. Because every stereoisomer has exactly one non-superimposable mirror image somewhere in that set of 8, they pair up into $8/2 = 4$ enantiomeric pairs. Any given member of a pair is a diastereomer (not an enantiomer) of the other six stereoisomers, since it differs in configuration at only some, not all, of the three stereocenters relative to them.

Q3 C CHEM/PHYS

Under S_N1 conditions (weakly nucleophilic, polar protic solvent, no strong base), the rate-determining step is unimolecular ionization of the carbon-leaving group bond to form a carbocation. Leaving-group ability tracks how well the departing group stabilizes its own negative charge and how weak the carbon-halogen bond is. Down the halogen column, C-X bond strength decreases and the resulting halide ion becomes larger and more able to spread out its negative charge, so iodide is both the weakest bond to break and the most stable ion once formed. Fluoride is the opposite on both counts: the C-F bond is the strongest carbon-halogen bond, and fluoride holds its negative charge tightly, making it a poor leaving group. The iodide substrate therefore ionizes, and reacts by S_N1 , much faster.

Q4 C BIO/BIOCHEM

Amino acids are classified by the chemistry of their side chains: nonpolar (hydrophobic), polar uncharged, acidic (negatively charged), and basic (positively charged). In an aqueous cytosol, folded globular proteins bury their nonpolar side chains in the interior and present polar and charged side chains to the water. Of the options, only leucine has a purely hydrocarbon side chain with no capacity to hydrogen-bond or carry charge, so it is the residue expected in the hydrophobic core; serine is polar (hydroxyl), aspartate is negatively charged (carboxylate), and lysine is positively charged (protonated amine) at physiological pH.

Q5 A BIO/BIOCHEM

Each level of protein structure has a characteristic set of interactions. Primary structure is the amino acid sequence joined by covalent peptide bonds; secondary structure consists of local, repeating backbone conformations — alpha helices and beta sheets — stabilized by hydrogen bonds between backbone carbonyl oxygens and amide hydrogens; tertiary structure is the overall three-dimensional fold of one chain, stabilized by side-chain interactions such as hydrophobic packing, salt bridges, and disulfide bonds. Because the question asks specifically about the alpha helix, the answer is backbone hydrogen bonding: side-chain-based interactions belong to higher structural levels, and peptide bonds belong to the primary level.

Q6 **C** PSYCH/SOC

Functionalism (structural functionalism) views society as a system of interdependent parts and explains social practices by the contributions they make to the stability of the whole. Arguing that a communal festival persists because it reaffirms shared values and strengthens social bonds is a classic functionalist account, tracing back to Durkheim's analysis of collective rituals producing social solidarity. Conflict theory would instead ask who gains power or resources from the arrangement; symbolic interactionism would examine how participants construct meanings during face-to-face encounters; and rational choice theory would model individual cost-benefit decisions, none of which matches the sociologist's stability-and-cohesion argument.

Q7 **A** PSYCH/SOC

Conflict theory, rooted in Marx's analysis of class struggle, explains social arrangements as outcomes of competition between groups with unequal power, in which dominant groups shape institutions to protect their own interests. The sociologist's argument has every element of this framework: a powerful group (wealthy property owners) used an institution (zoning law) to impose costs (pollution) on a less powerful group (poor residents), and the arrangement endures because it continues to serve the powerful. Functionalism would explain persistence through society-wide benefits, symbolic interactionism would examine meanings in daily interaction, and social exchange theory would analyze reciprocal costs and rewards between individuals — none fits an account centered on institutionalized group advantage.

Q8 **C** CARS

The passage sets up the conventional refinement narrative, presents the revisionist counter-narrative that silence was manufactured, and then qualifies the revisionist case with two pieces of evidence it cannot absorb: the demands of longer symphonic works and the voluntary adoption of silence by workingmen's concert societies. The closing paragraph draws the actual thesis, which is that norms of attention are themselves historical artifacts, shaped by both deliberate discipline and a changing repertory. Only the answer stating that audience conventions are historical products formed by both forces captures this synthesis; the other options either restate the refinement view the author questions, absolutize the revisionist view the author explicitly limits, or inflate a single detail into the main claim.

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